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Project Name: Student Exam Performance Database Analysis

Department: Computer Science & Engineering

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Dataset Description

Source of the dataset and the real-world problem it relates to

The dataset titled "Student Performance Factors" and is used to analyze the various factor that influence student's academic performance.

Source: Kaggle

Total number of observations and variables/features

Number of observations (rows): 6607
Number of features (columns): 20

Feature Description

No.	Feature Name	Type	Category
1	Hours_Studied	Numerical	Continuous
2	Attendance	Numerical	Continuous
3	Sleep_Hours	Numerical	Continuous
4	Tutoring_Sessions	Numerical	Discrete
5	Parental_Involvement	Categorical	Ordinal
6	Access_to_Resources	Categorical	Nominal
7	Extracurricular_Activities	Categorical	Binary
8	Motivation_Level	Categorical	Ordinal
9	Internet_Access	Categorical	Binary
10	Family_Income	Categorical	Ordinal
11	Teacher_Quality	Categorical	Ordinal
12	School_Type	Categorical	Nominal
13	Peer_Influence	Categorical	Ordinal
14	Learning_Disabilities	Categorical	Binary
15	Parental_Education_Level	Categorical	Ordinal
16	Distance_from_Home	Numerical	Continuous
17	Exam_Score	Numerical	Continuous
18	Previous_Scores	Numerical	Continuous
19	Physical_Activity	Numerical	Discrete
20	Gender	Categorical	Nominal

Table-01

Central Tendency and Dispersion Analysis

No.	Numeric Features	Mean	Median	Mode	Variance	Std Dev	IQR
1	Hours_Studied	19.975329	20.0	20	35.887221	5.990594	8.0
2	Attendance	79.977448	80.0	67	133.344178	11.547475	20.0
3	Sleep_Hours	7.029060	7.0	7	2.155377	1.468120	2.0
4	Previous_Scores	75.070531	75.0	66	207.353789	14.399784	25.0
5	Tutoring_Sessions	1.493719	1.0	1	1.514304	1.230570	1.0
6	Physical_Activity	2.967610	3.0	3	1.063438	1.031231	2.0
7	Exam_Score	67.235659	67.0	68	15.135646	3.890456	4.0

Table-02

From the *Table-02* ,

1. **Hours_Studied:** The Mean(19.97),Median(20.0), and Mode(20.0) are nearly identical , indicating a **symetric distribution**. The Standard Derivation is 5.99, and the coefficient of variation (CV= (SD/Mean)*100) is approximately 29.9899% of the mean.This indicates **moderate varibility** suggesting that the data are somewhat spread out around the mean , with noticeable variable among observations. Additionally, the IQR (8.0) reflects a moderate spread without the presence of extreme variability.

2. **Attendance:** Altought the Mean(79.97) and Median(80.0) are close, but Mode(67.0) differs , indicating a **slight skew** at a lower value. The Standard Derivation is 11.54, and the coefficient of variation (CV= (SD/Mean)*100) is approximately 14.43% of the mean.This indicates **low varibility** suggesting that

the data are fairly close to the mean. Additionally, the IQR (20.0) reflects a indicate **significant variability** in attendance.

3. **Sleep_Hours:** The Mean(7.029),Median(7.0), and Mode(7.0) are nearly identical , indicating a **symetric distribution**. The Standard Derivation is 1.46, and the coefficient of variation is approximately 20.77% of the mean.This indicates **moderate varibility** suggesting that the data are somewhat spread out around the mean , with noticeable variable among observations. Additionally, the IQR (2.0) showing **closely clustered** in sleep patterns.

4. **Previous_Scores:** Altought the Mean(75.07) and Median(75.0) are close, but Mode(66.0) is lower , indicating a **slight left skew**. The Standard Derivation is 14.39, and the coefficient of variation is approximately 19.17% of the mean.This indicates **low varibility** suggesting that the data are fairly close to the mean. Additionally, the IQR (25.0) reflects a indicates a wide spread.

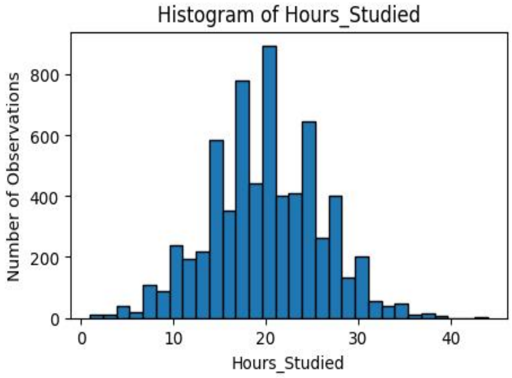
5. **Tutoring_Sessions:** The mean(1.49) is slightly higher than median(1.0) and mode(1.0), indicating a **right-skewed distribution**. The Standard Derivation is 1.23, and the coefficient of variation is approximately 82.55% indicating, **relatively high varibility**. This suggests that while most students attend few sessions, some attend significanlty more. The IQR(1.0) shows that the majority of students fall within a narrow range.

6. **Physical_Activity:** The Mean(2.9676),Median(3.0), and Mode(3.0) are nearly identical , indicating a **symetric distribution**. The Standard Derivation is 1.031, and the coefficient of variation is approximately 34.74% suggesting **relatively high varibility.** however, the IQR (2.0) reflects that most values are still **moderately clustered**.

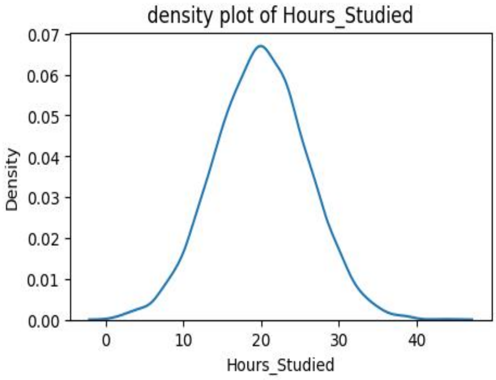
7. **Exam_Score:** The mean(67.23), median(67.0), and mode(68.0) are very close, indicating a **nearly symmetric distribution**. The standard deviation(3.89,)and the coefficient of variation is approximately 5.786% indicating very low variability meaning that exam scores(data) **tightly clustered around mean** and IQR (4.0) also support this.

Distribution and Shape

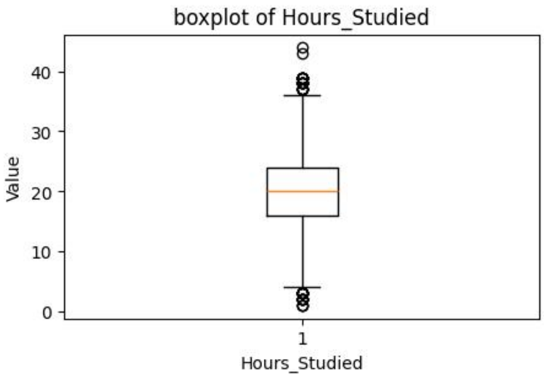
1. Hours_Studied:



The histogram of Hours_Studied shows how frequently different study durations occur. The distribution appears slightly concentrated around the middle range, indicating that most students study a moderate number of hours.



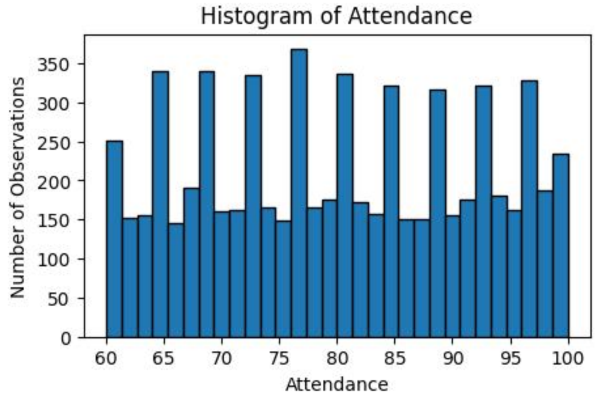
The smooth curve looks balanced on both sides, which means the data is evenly spread.



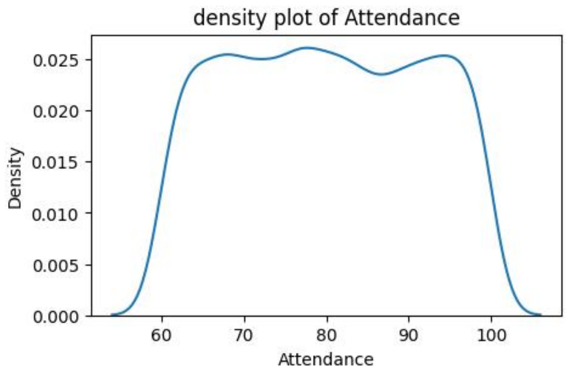
The boxplot shows that most values are close to each other. There are only a few unusual values.

The data is **almost symmetric** (balanced). This means most students study around the average. There are a few outliers, but they are not very extreme.

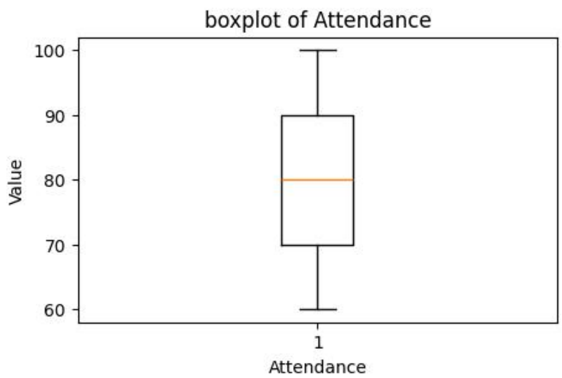
2. Attendence:



The histogram shows that most students have high attendance.



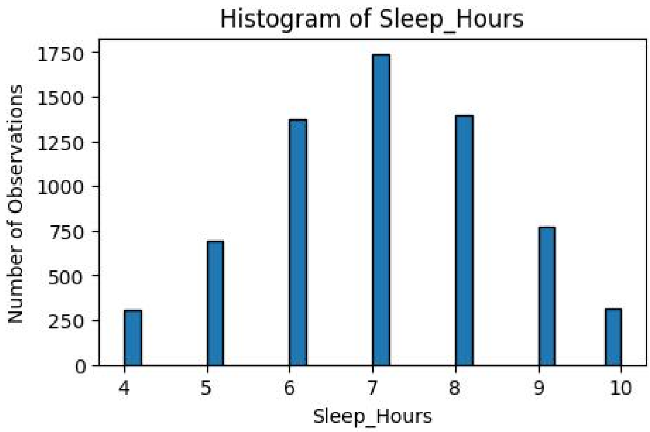
The curve leans to one side, showing that only a few students have low attendance.



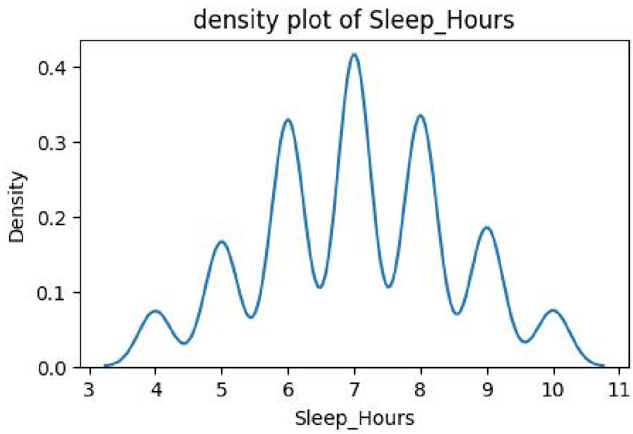
The boxplot shows some low values that are far from the rest.

The data is left-skewed, meaning most students attend classes regularly, but a few do not. Some low attendance values are outliers.

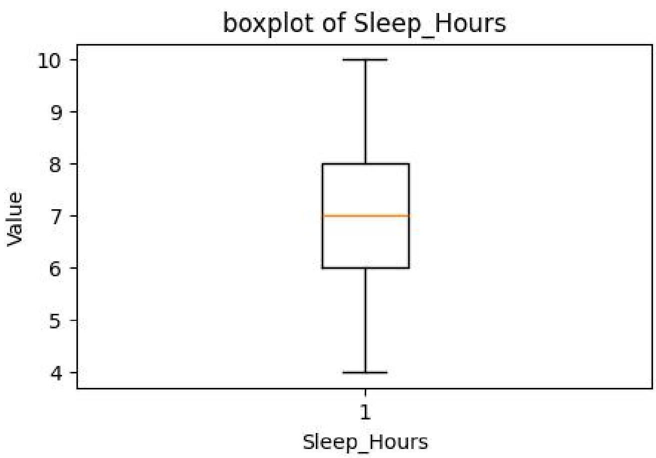
3. Sleep Hours:



Most students sleep around a normal number of hours.



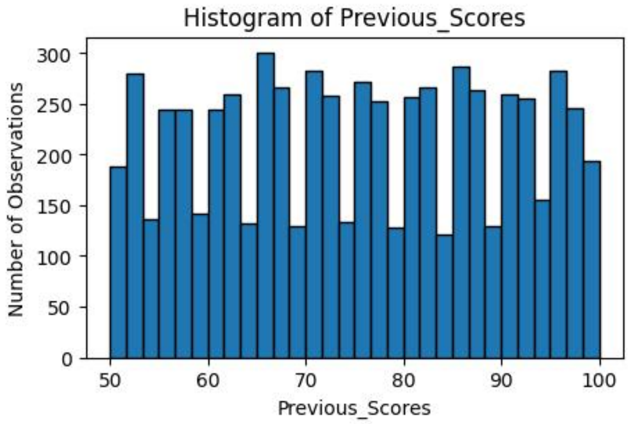
The curve looks balanced, showing no strong skew.



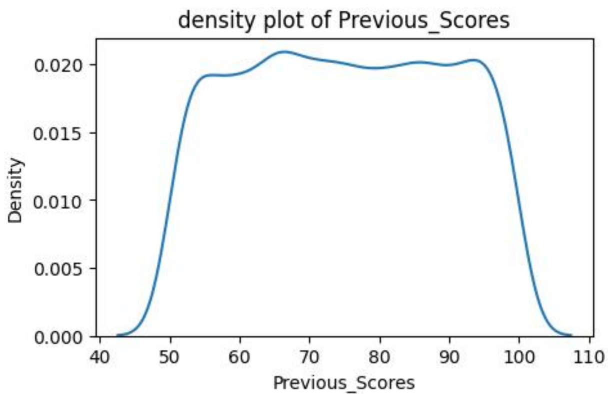
The values are close together with very few extreme values.

The data is **symmetric**, meaning sleep hours are evenly spread. No outliers are present.

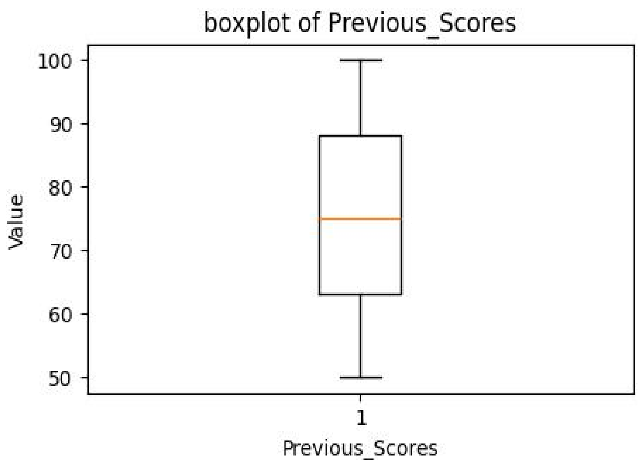
4. Previous Score:



Most students have medium to high previous scores.



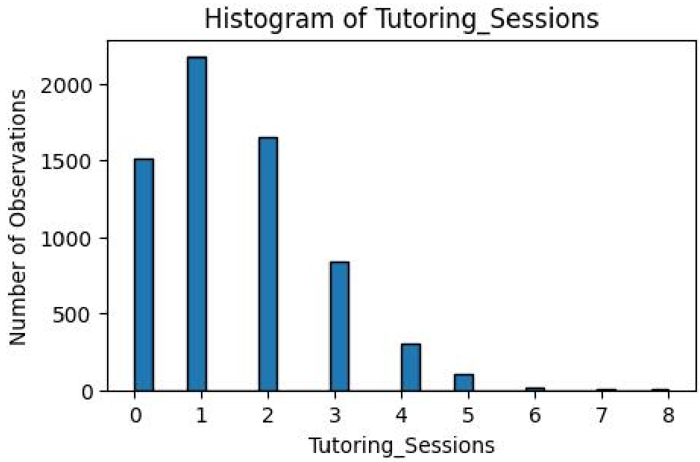
The curve shows a slight stretch toward higher values.



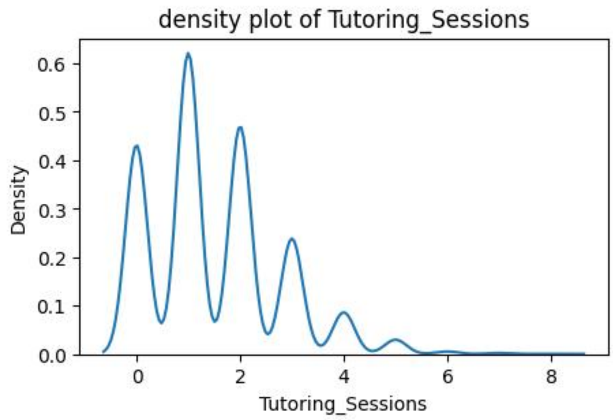
Some higher scores are a bit far from the rest.

The data is slightly **right-skewed**, meaning a few students scored much higher than others. A few high scores are outliers.

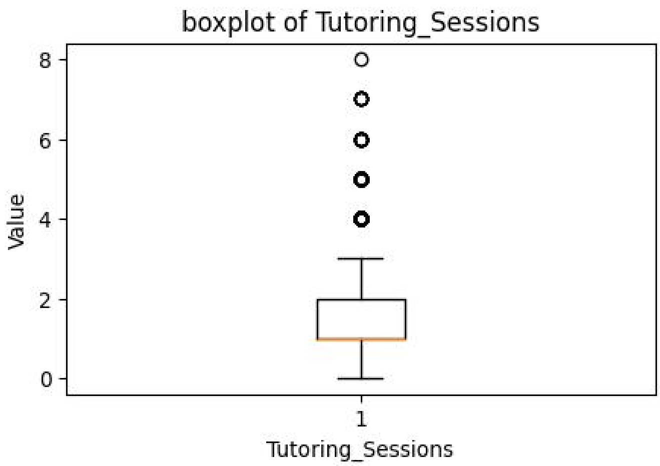
5. Tutoring Sessions:



The histogram is **Right-Skewed**. Most students attend between 0 and 2 sessions, with the frequency dropping sharply as the number of sessions increases.

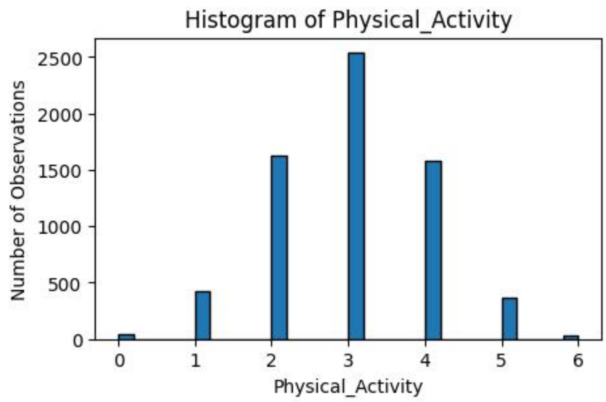


The peak is concentrated at the lower end (left side), confirming that the majority of the population does not attend frequent tutoring.

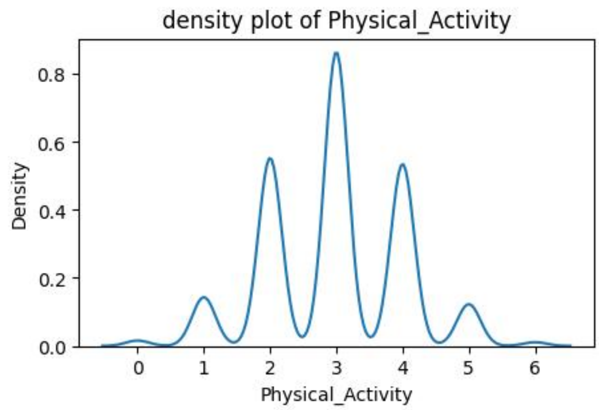


This boxplot identifies **430 outliers**. These represent a specific group of students who attend a much higher number of sessions than the typical student.

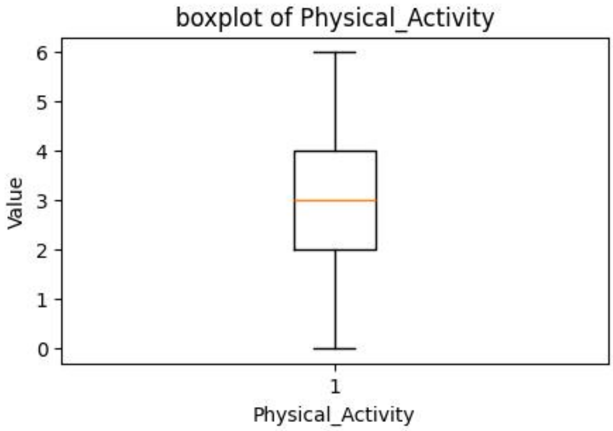
6. Physical Activities:



The histogram shows a symmetric distribution centered around 3 hours. The bars are evenly spread, showing that most students engage in a moderate amount of physical exercise per week.

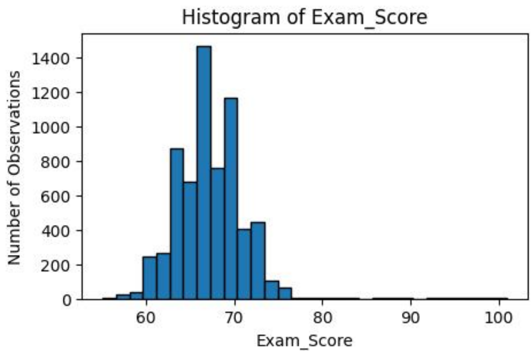


The density curve is bell-shaped and highly symmetric (Skewness: -0.03), indicating that the physical activity levels of students follow a normal distribution without any heavy leaning toward the high or low ends.

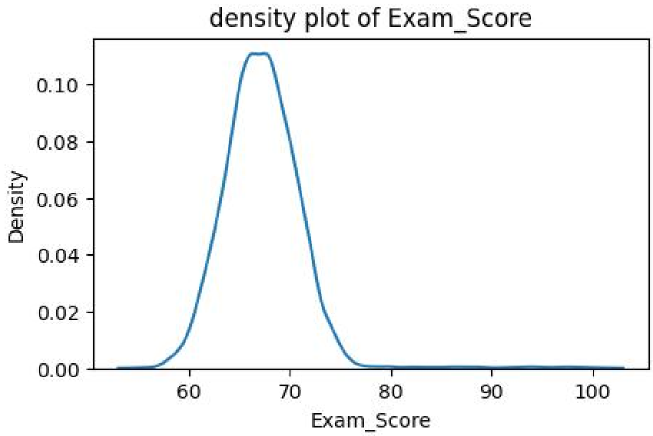


The boxplot is well-balanced with a median line in the center of the box. There are **0 outliers**, confirming that all students' physical activity levels fall within a standard, expected range for this group.

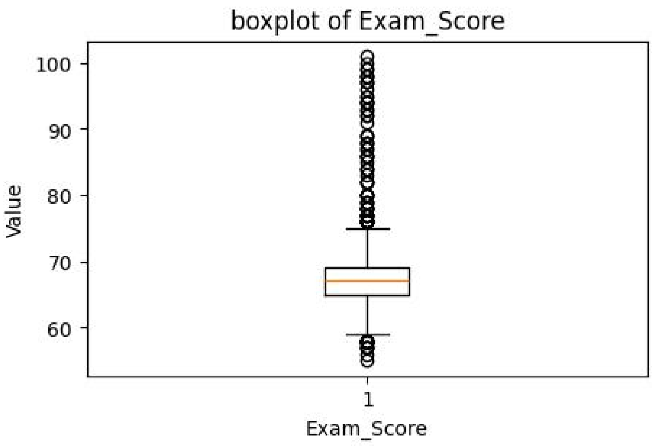
7. Exam Score:



The histogram is visibly **Right-Skewed**. There is a large concentration of students scoring between 60 and 70, with a long tail extending toward higher scores.

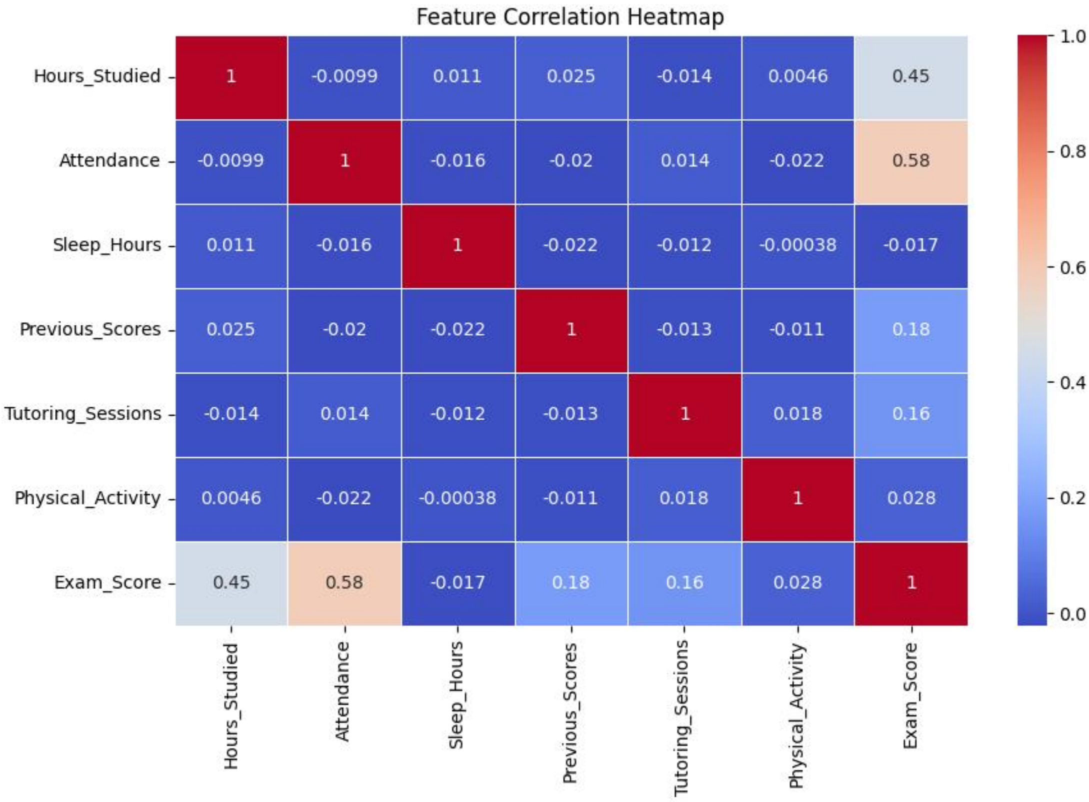


The density plot confirms the skewness (1.64), showing that achieving a very high score is less common and belongs to a smaller "tail" of students.



The boxplot shows **104 outliers** on the upper side. These are your "High Achievers" who scored significantly above the median range of the class.

Heatmap



The Heatmap shows the correlation between numeric columns.

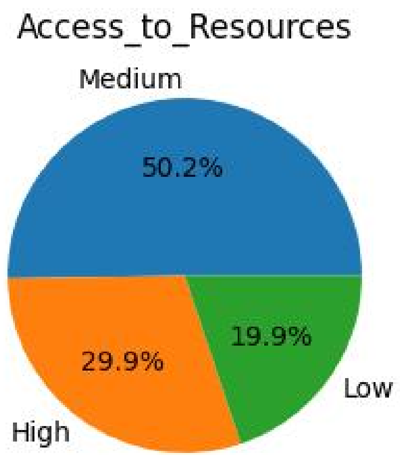
Summary Table

Numeric Features	Distribution Shape	Skewness	Outliers Found
Attendance	Symmetric	0.01	0
ExamScore	Right Skewed	1.64	104
Hours Studied	Symmetric	0.013	43
Previous Scores	Symmetric	-0.003	0
Sleep Hours	Symmetric	-0.023	0
Tutoring Sessions	Right Skewed	0.81	430

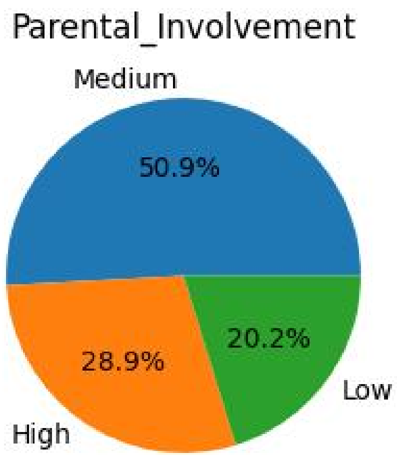
Categorical Variable Analysis

In this section, we analyze the non-numerical factors. Most categorical variables in this data sets are well-distributed, but some show significant imbalance.

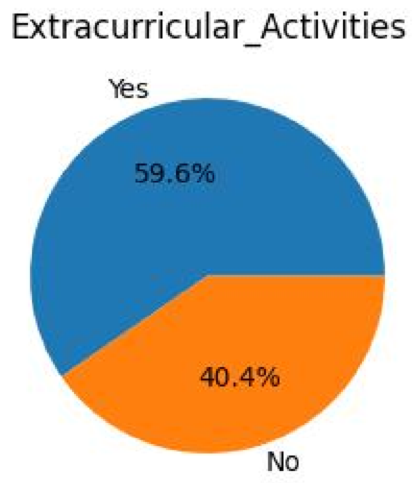
1. Access to Recources:



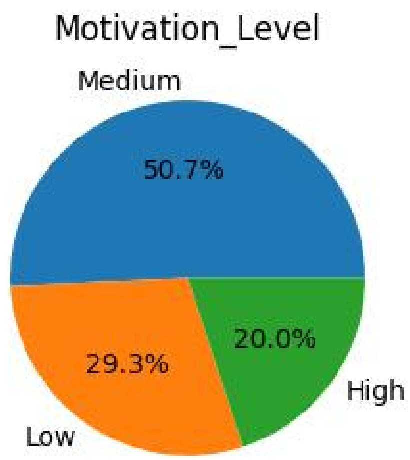
2. Parental involvement:



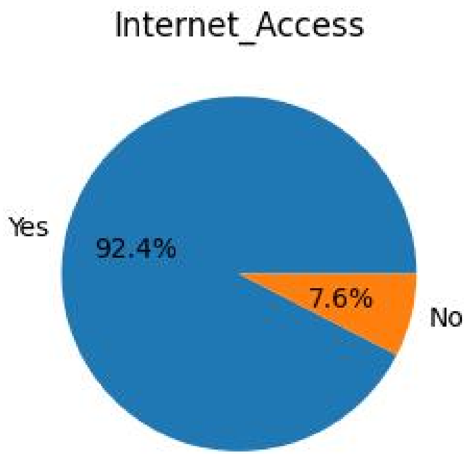
3. Extracurricular Activities:



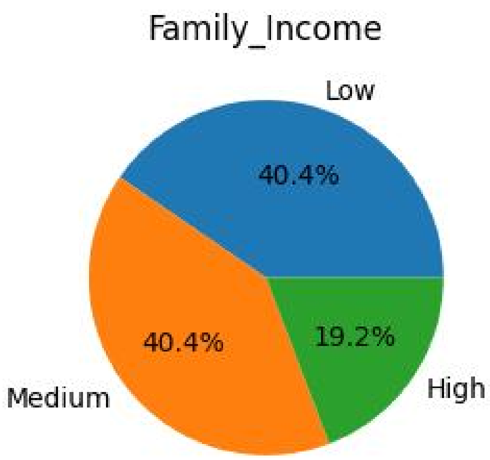
4. Motivational Level:



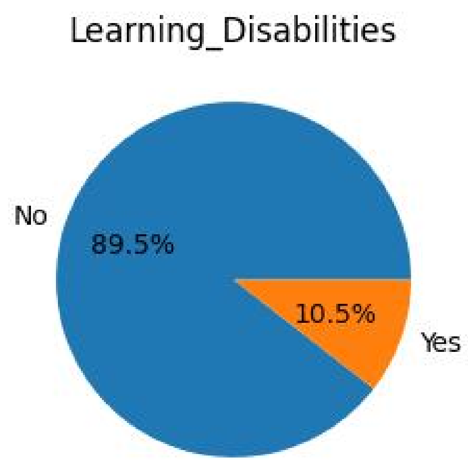
5. Internet Access:



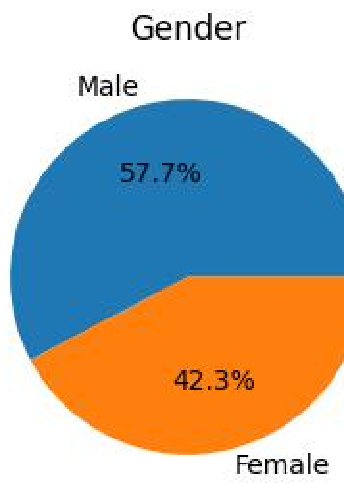
6. Family Income:



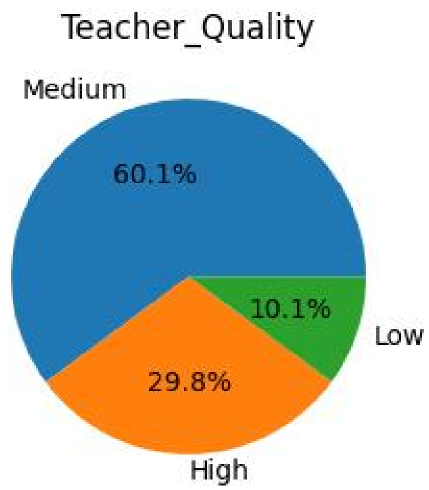
7. Learning Disabilities:



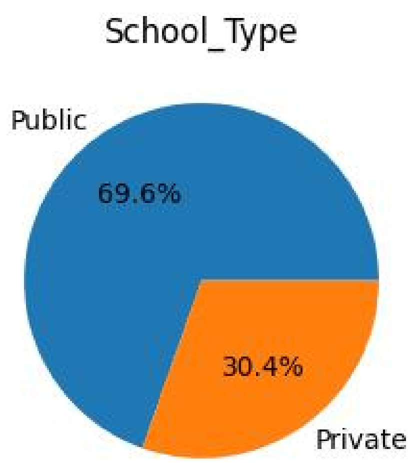
8. Gender:



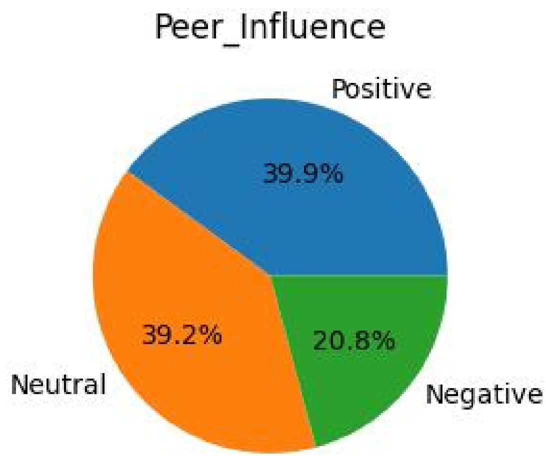
9. Teacher Quality:



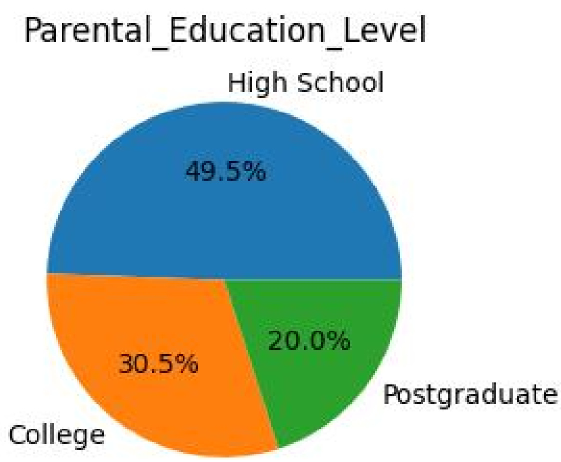
10. School Type:



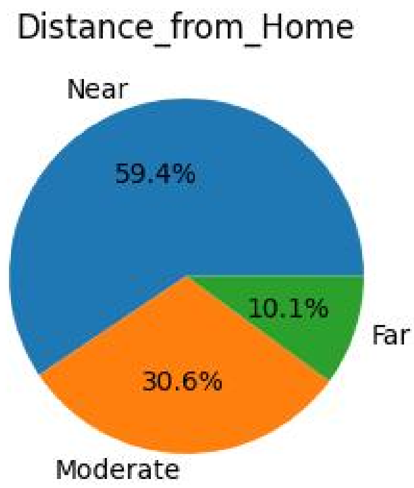
11. Peer Influence:



12. Parental Education Level:



13. Distance From Home:



Frequency Table: Key Categorical Variables

Features	Most Common Category	Frequency	Percentage
Internet Access	**Yes**	**6108**	**92.4%**
Parental Involvemnet	**Medium**	**3362**	**50.9%**
Motivational Level	**Medium**	**3358**	**50.7%**
Gender	**Female**	**3334**	**50.5%**
Extracurricular Acts	**Yes**	**3938**	**59.6%**

Observations:

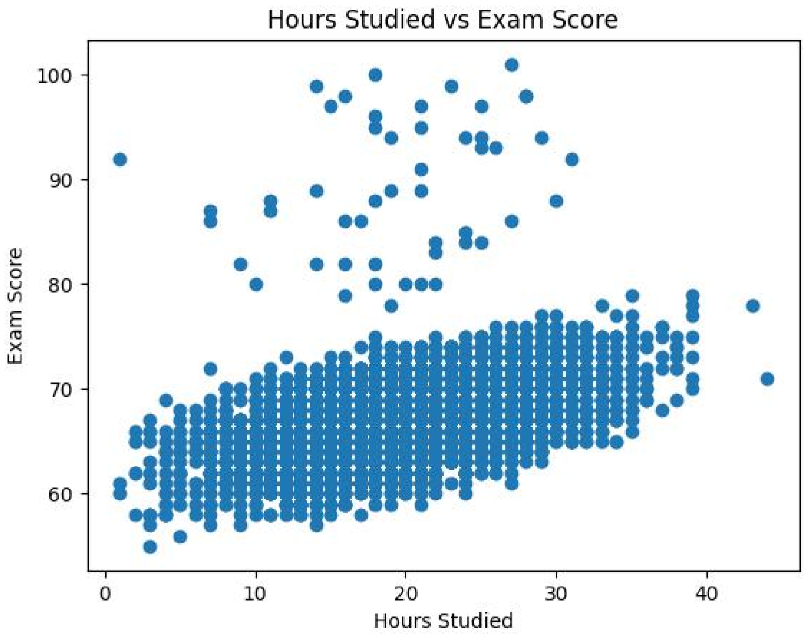
- **Common Categories:** The most frequent category across behavioral factors (Involvement, Motivation) is "Medium". For resources,"Learning Disabilities(No)" ,"Public School" and "Internet Access (Yes)" are most common.

- **Imbalance Notes:** There is a heavy imbalance in **Internet Access** (92.45% have it). This could make it difficult to generalize the impact of not having internet due to the small sample size in that category. **Learning Disabilities** also shows an imbalance, with 89.48% of students having none.

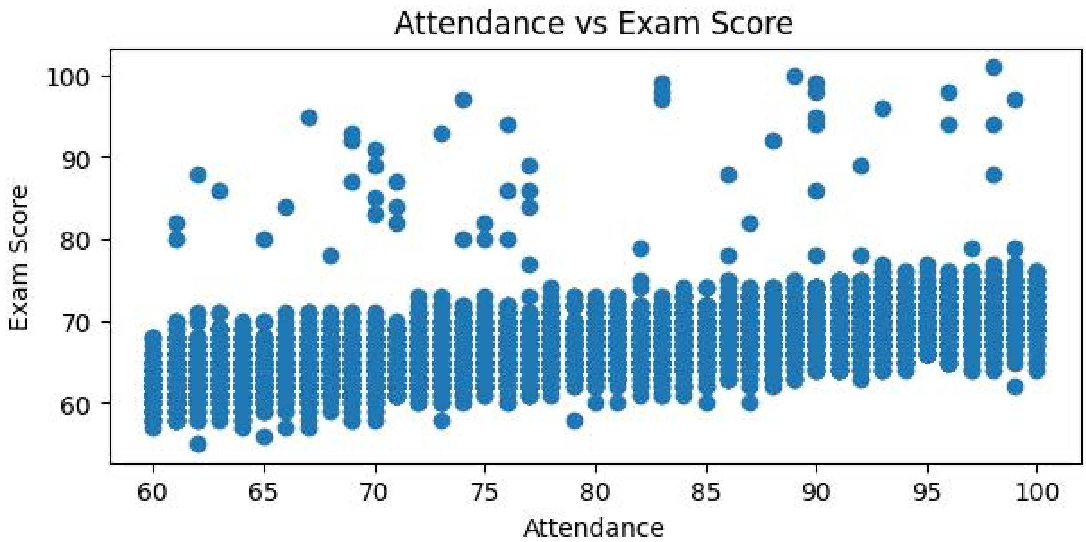
Relationships Between Variables

We analyzed the relationships between numerical variables, specifically focusing on how they correlate with the **Exam Score**.

Hours Studied VS Exam Scores

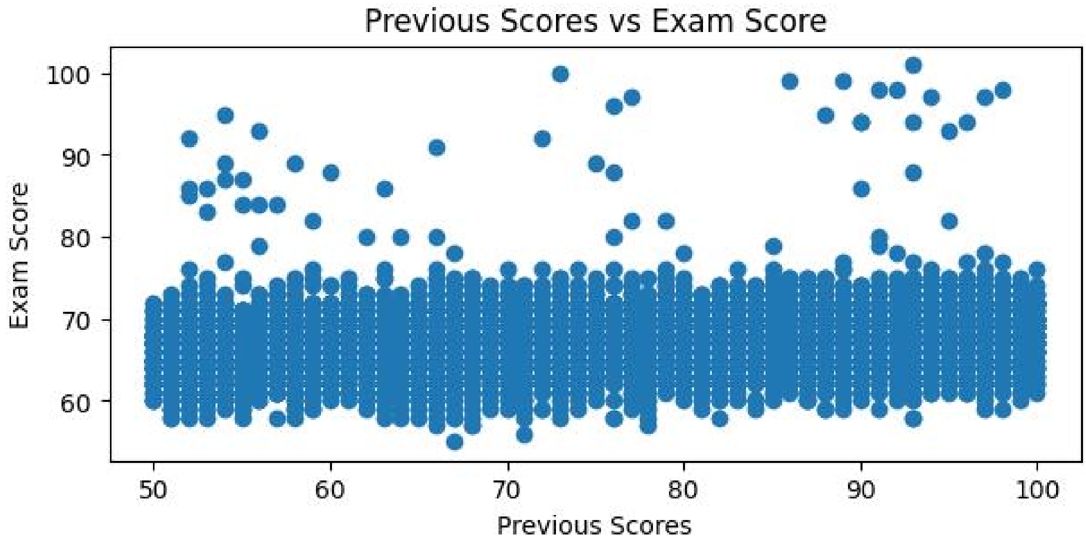


A positive trend is visible; however, there is significant spread. This suggests that while more study hours help, the effectiveness of that study varies across the student population.



There is a strong linear relationship between attendance and scores. Students with higher attendance percentages consistently achieve higher marks, making it a critical predictor for the final report.

Previous Scores VS Exam Scores



Surprisingly, the relationship between previous scores and the final exam is less defined than attendance. While a positive trend exists, many students with lower previous scores still performed well on the final exam.

Data Quality Assessment

A data audit was performed to ensure the integrity and reliability of the analysis.

- **Missing Values:** Small gaps were found in Teacher Quality (78), Parental Education Level (90), and Distance from Home (67). Representing less than 1.5% of the 6,607 records, these are statistically insignificant and can be handled via Mode Imputation (replacing nulls with the most frequent category).

- **Duplicate Rows:** The dataset contains **0 duplicates**, confirming that each record is a unique student observation.

- **Unusual Values (Outliers):** Using the **IQR Rule**, 104 outliers were identified in Exam_Score. These represent "high-achievers" rather than errors; they provide critical insight into the factors driving top-tier academic performance and were therefore retained in the study.

Feature Engineering & Data Preparation

Two engineered features were added to capture interaction effects:

- study_efficiency = Previous_Scores / (Hours_Studied + 1) — measures score yield per study hours
- engagement = Attendance × Hours_Studied — combined effort metric

Train/test split: 70/30 (stratified, random_state=42). Outliers in Hours_Studied, Attendance, and Previous_Scores were removed from BOTH the training set (4,596 rows) and the test set (1,983 rows). Outliers in Hours_Studied, Attendance, and Previous_Scores were removed from the training set only → final train size: 4,596 rows. Test set (1,983 rows) was kept untouched to simulate real-world evaluation.

Linear Regression Model

The first column of ones has been added to matrix A to address the problem of bias. The method of Moore-Penrose pseudo-inverse was used for stability purposes (np.linalg.pinv). The learning rate for Gradient Descent is α=0.01 and number of iterations = 1000.

Regression Coefficients & Interpretation

Feature	Coeff. (NE)	Coeff. (sklearn)	Coeff. (GD scaled)	Interpretation
Intercept	43.25	43.25	67.22	Baseline score
Hours_Studied	0.22	0.22	0.59	+0.22 per extra hour
Attendance	0.18	0.18	1.77	+0.18 per 1% attendance
Previous_Scores	0.05	0.05	0.70	Weak positive effect
study_efficiency	0.01	0.01	-0.02	Minimal contribution
engagement	0.00	0.00	1.19	Negligible direct effect

Note: Gradient Descent coefficients are on the normalized scale and not directly comparable in magnitude to Normal Equation coefficients. Both methods converge to the same predictions.

Model Performance

Model	MSE	RMSE	Note
Normal Equation (numerical only)	5.21	2.28	Baseline
sklearn LinearRegression	5.21	2.28	Confirms NE
Gradient Descent (normalized)	5.23	2.29	Converges to NE
Normal Equation + Categorical	3.46	1.86	Best model

- Normal Equation and sklearn produce identical results (RMSE 2.28) — confirming the manual implementation is correct.
- Gradient Descent converges to the same solution (RMSE 2.29) — the 0.01 gap is negligible.
- Adding one-hot encoded categorical features (School_Type, Gender, Peer_Influence, etc.) reduces RMSE from 2.28 → 1.86, an 18% improvement.
- Given Exam_Score std = 3.89, an RMSE of 1.86 means predictions are off by less than half a standard deviation on average — a strong result.

Visualizations

Figure 1: Actual vs. Predicted Exam Score (Test Set)

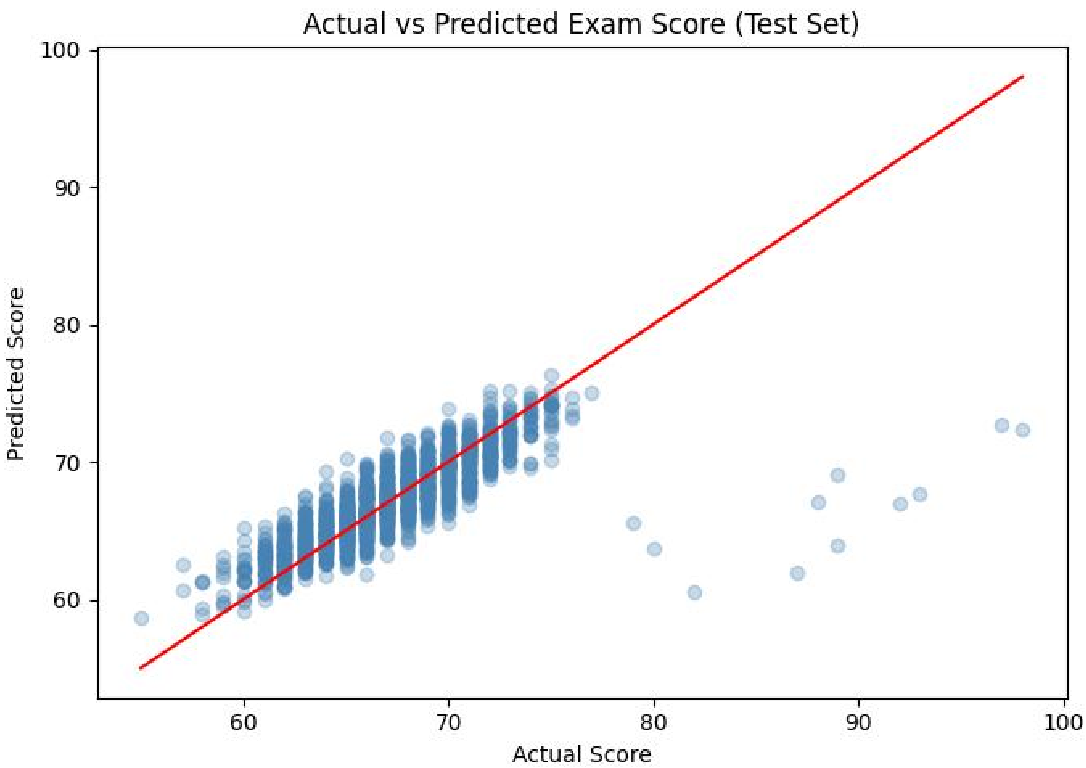


Figure 1: Each point represents a test-set student. The red diagonal is the perfect-prediction line. The tight cluster around it confirms RMSE ≈ 2.28 — the model predicts most scores within 2–3 points.

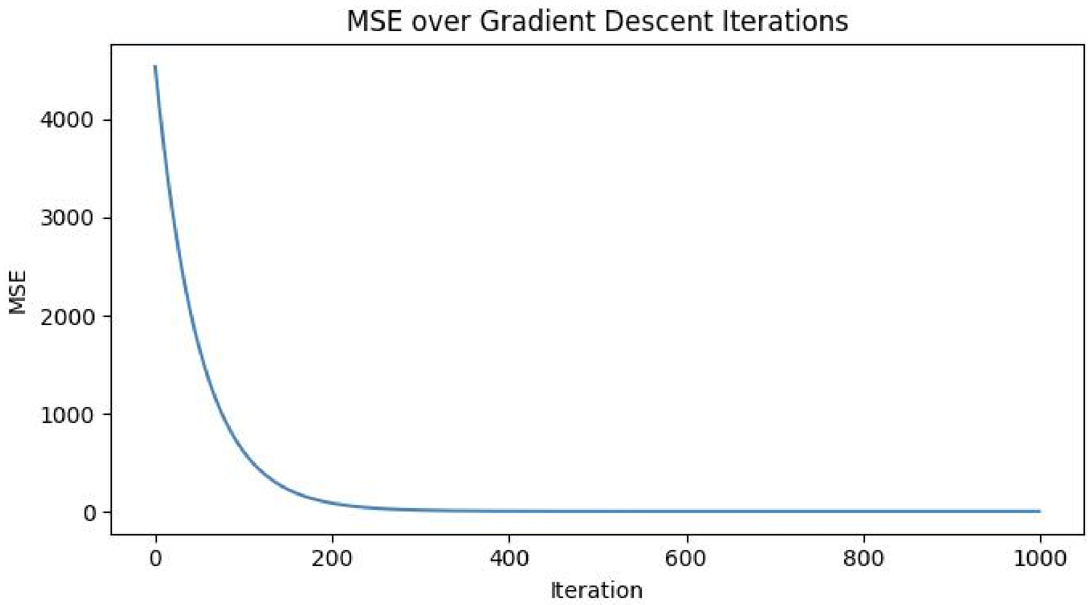


Figure 2: MSE drops steeply in early iterations and plateaus near iteration ~300, confirming learning rate α = 0.01 is well-chosen and the model converges to the Normal Equation solution

Key Insights

After looking at all the charts and numbers, here are the main things we learned about what makes a student successful:

1. **Attendance is the #1 Factor:** Out of everything we looked at, Attendance had the biggest impact on grades. The more a student shows up to class, the better their exam score tends to be. It's the most consistent way to predict if a student will do well.

2. **Most Students Have Similar Habits:** For things like Sleep and Study Hours, most students are pretty much in the middle. The data looks like a classic bell shape, which tells us that extreme habits (like studying 40 hours or never sleeping) are actually quite rare.

3. **The Advantage of Extra Support:** While almost everyone has the internet, the students who really stand out are the ones with High Parental Involvement. These

students are much more likely to be in that "top-performer" group we saw in the outliers.

4. **The Big Three:** If we wanted to guess a student's grade in the future, the three most important things to check would be their Attendance, their Study Hours, and how involved their Parents are. These three give us the best idea of how a student is going to perform.